EIN 4343 Inventory and Supply Chain Systems

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The role of transportation

- Movement of product from one location to another
- Products rarely produced and consumed in the same location
- Two agents
  - Shipper requires the movement of the product
  - Carrier moves or transports the product

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Transportation systems

- Wider market reach
- Economies of scale
- Reduced product prices

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Logistics costs reshaping US supply chains

*The U.S. logistics sector just crossed a milestone that few saw coming — **$2.6 trillion in total logistics costs** for 2024, equal to nearly **9% of U.S. GDP**.*

*That figure, revealed in the latest State of Logistics Report, represents not only record growth but a **structural shift** in how America moves goods, manages inventory, and prices freight.*

*From global imports to final-mile deliveries, **every cost line — fuel, labor, warehousing, insurance, compliance, and technology — has surged**.*

*But hidden inside that inflation is a transformation: carriers, shippers, and 3PLs are rewriting the cost equation through **digitization, network redesign, and data-driven optimization**.*

*The $2.6 trillion number is not a ceiling — it's a signal that the U.S. logistics model itself is evolving from volume-driven to **value-driven**.*

https://amblogistic.us/how-logistics-costs-are-reshaping-american-supply-chains/

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Why $2.6 trillion?

Fuel and Energy Volatility

*Diesel averaged **$4.19 per gallon** in 2024, up nearly 16% from the previous year. For carriers, fuel now accounts for **25–30% of total operating expenses**. Many fleets are mitigating through **hedging strategies, route optimization, and alternate fuels**, but volatility is here to stay.*

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Why $2.6 trillion?

Labor Shortages and Wage Escalation

*The U.S. still faces a shortfall of **80,000 truck drivers**. Wages rose 8–10% in 2024, and retention costs are climbing faster than recruitment budgets.*
Automation can't fully replace skilled drivers — but digital load assignment, telematics support, and flexible scheduling are softening the hit.

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Why $2.6 trillion?

Warehousing Costs Reach Historic Highs

*Average industrial rent in key markets like Dallas, Atlanta, and Inland Empire jumped **12% year-over-year**.*

*E-commerce demand and inventory buffers post-COVID have pushed warehouse utilization near capacity, leading to creative solutions: **micro-fulfillment centers, AI slotting, and multi-client storage models.***

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Why $2.6 trillion?

Insurance and Compliance Inflation

Cargo theft, cyberattacks, and litigation risks have doubled insurance premiums in some corridors.

Meanwhile, ESG and emissions compliance add cost layers — but they're fast becoming competitive differentiators.

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Why $2.6 trillion?

Technology as a Cost and a Cure

*AI-based Truck Management Systems, Internet-of-Things telematics, and predictive analytics represent upfront expense — yet they deliver efficiency returns of **10–25%** within the first year of deployment.*

https://amblogistic.us/how-logistics-costs-are-reshaping-american-supply-chains/

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Relevant costs in transportation

- Fixed transportation costs
  - Roadway acquisition and maintenance, terminal operations, transportation equipment, carrier administration
- Variable transportation costs
  - Fuel, labor, equipment maintenance, handling, pickup and delivery
- In-transit inventory costs
  - Opportunity cost while in transit

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Modes of transportation

- Air
- Package carriers
- Truck
- Rail
- Water
- Pipeline
- Intermodal

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Modes of transportation

	Delivery lead time	Cost structures	Product characteristics
Air	Fast	High infrastructure High shipment costs	Small, high-value, time-sensitive
Truck – TL – LTL	Fast	High infrastructure; Low equipment; Low shipment	Any
Rail	Slow	High infrastructure; High equipment; Low shipment	Large, heavy, high density, low value, time insensitive
Water	Slow	High infrastructure; High equipment; Low shipment costs	Large loads
Pipeline	Slow	High infrastructure	Crude oil; natural gas; refined oil products
Intermodal – Water/rail – Truck/water/rail		Existing port, airport, railway, interstate structures	Any

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Why are Nike trainers washing up on beaches?

- Flores Island, Azores Archipelago - September 2018
- Cornwall, UK – May 2019

https://www.bbc.com/news/uk-48464664

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Where the products have been found

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Why are Nike trainers washing up on beaches?

"*The source of all these shoes is believed to be a single ship.*

Mr Ribeiro said: "Through the research I have done everything indicates they may have been from some of the 70 to 76 containers that fell overboard from the Maersk Shanghai."

In early spring last year, the Maersk Shanghai – a 324m (1,063ft) vessel capable of carrying more than 10,000 shipping containers – was travelling from Norfolk, Virginia, down the east coast of the US to Charleston, South Carolina.

On the evening of 3 March – 17 miles from the Oregon Inlet, off the coast of North Carolina – it was caught in a storm. While battling high winds and rough seas, a stack of its cargo-laden containers toppled overboard."

https://www.bbc.com/news/uk-48464664

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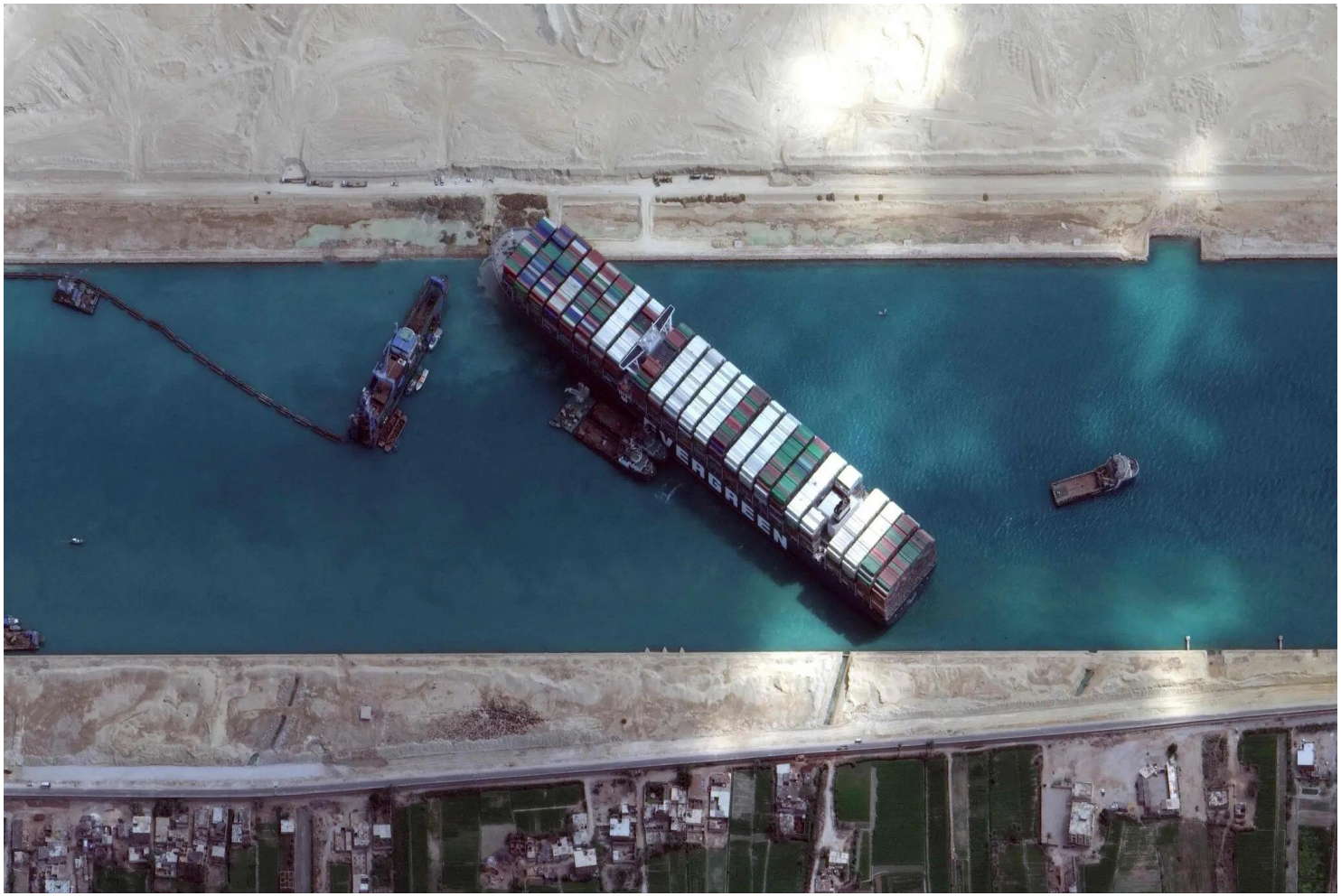
Ever Given



https://www.nytimes.com/2021/07/17/world/middleeast/suez-canal-stuck-ship-ever-given.html/

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How one of the world's biggest ships jammed the Suez Canal

"When the Ever Given — one of the largest container ships ever built, more sideways skyscraper than boat — got stuck in the Suez Canal for six days in March, it dammed up worldwide shipping and froze nearly $10 billion in trade a day.…"

"While dozens of other freighters roughly the size of the Ever Given had traversed the Suez in the previous year without problems, shipping analysts have been warning for years that container ships have grown too large to safely operate in many ports and canals."

https://www.nytimes.com/2021/07/17/world/middleeast/suez-canal-stuck-ship-ever-given.html/

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The Ambassador Bridge

Truck blockade at US-Canadian border shuts auto plants

“ General Motors is the latest automaker to suspend production because of protests at a crucial Canadian border crossing that have disrupted supply chains already in turmoil because of the pandemic.

G.M. said it had canceled two shifts on Wednesday and Thursday at a factory in Lansing, Mich., that makes sport utility vehicles. The plant depends on components that normally travel across a bridge between Windsor, Ontario, and Detroit that was closed by truck drivers and far-right groups angry about vaccine mandates and demanding the resignation of Prime Minister Justin Trudeau.

Ford Motor and Toyota have also shut down some operations because factories could not get parts manufactured in Canada.

The shutdown at the border will prevent manufacturing at Toyota’s three Canadian plants for the rest of this week, Scott Vazin, a spokesman for Toyota Motor North America, said Thursday.”

“As long as the shutdowns are short lived, the impact on automaker sales and worker livelihoods should be limited. Companies are likely to make up for any lost production by running extra shifts or other measures.”

https://www.nytimes.com/live/2022/02/10/world/protests-in-canada

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Transportation management decisions

- Strategic
  - Network design
- Tactical
  - Mode selection
- Operational
  - How frequently to ship
  - How much to ship
  - How to route

Inventory management problem

Vehicle routing problem

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Design options for a transportation network

- When designing a transportation network
  1. Should transportation be direct or through an intermediate site?
  2. Should the intermediate site stock product or only serve as a cross-docking location?
  3. Should each delivery route supply a single destination or multiple destinations (milk run)?

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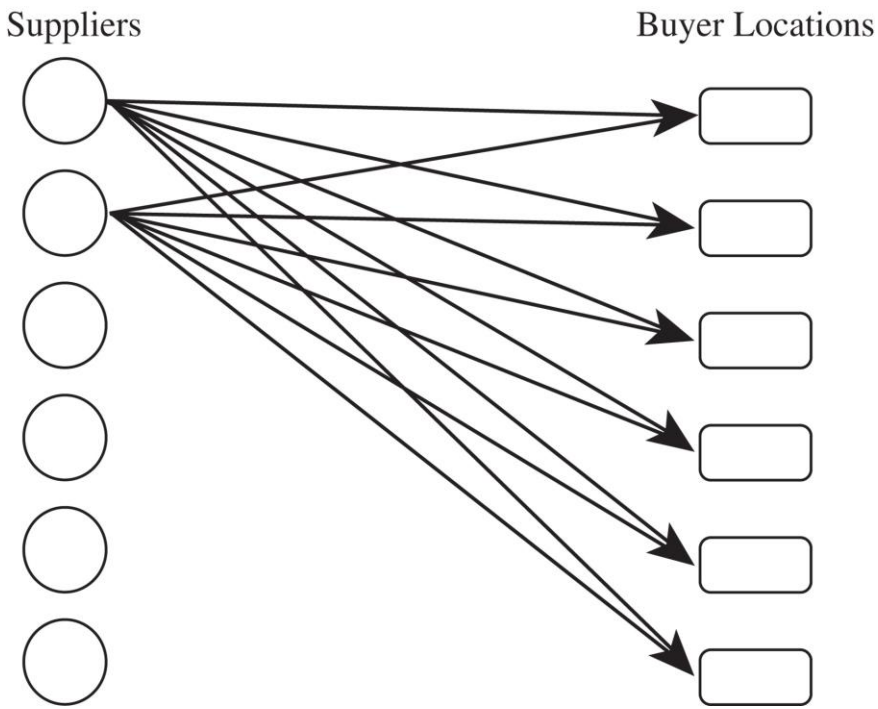
Direct shipment network

Disadvantages:
High inventories (due to large lot size)
Significant receiving expense



Advantages:
Easy to coordinate.
No intermediate storage or handling.

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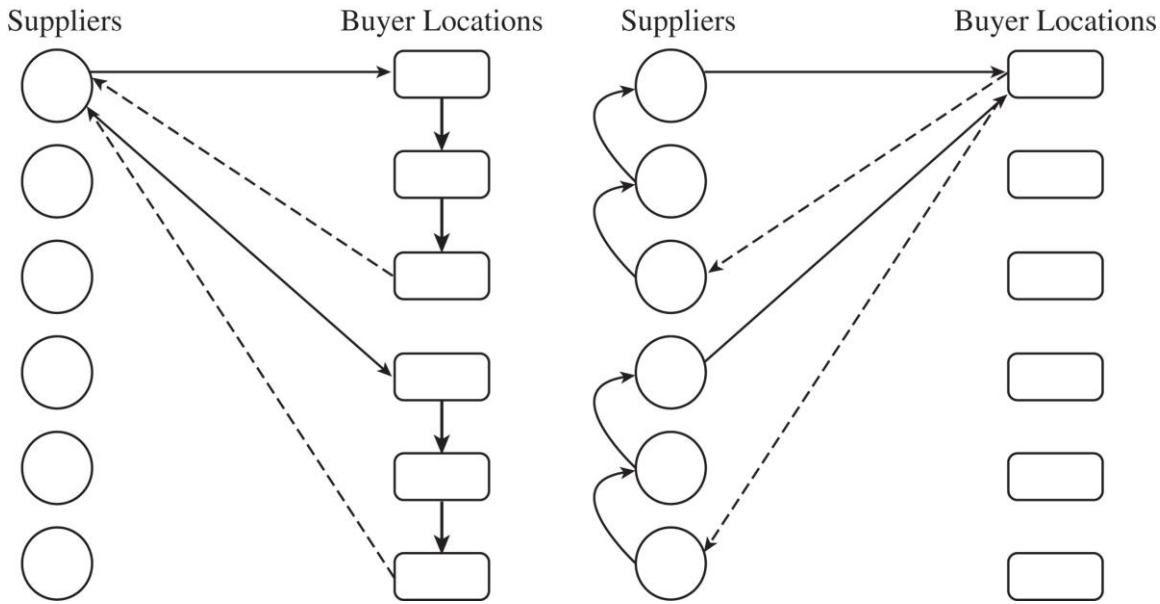
Direct shipping with milk runs

Disadvantages:
Increased coordination complexity



Advantages:
Lower transportation costs for small lots
Lower inventories

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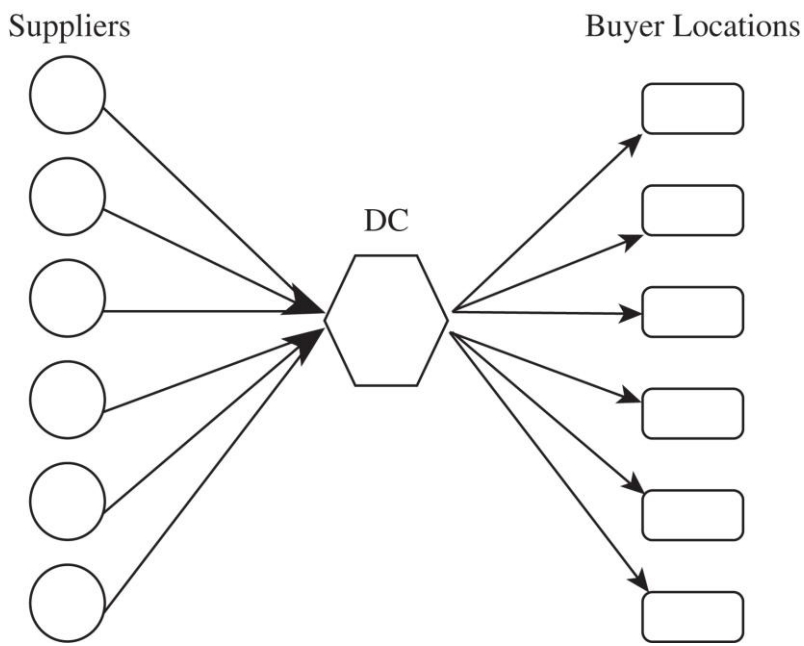
All shipments via intermediate distribution center (DC) with storage

Disadvantages:
Increased inventory cost
Increased handling at DC



Advantages:
Lower inbound transportation cost through consolidation

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Crossdocking

https://www.supplychain247.com/article/capitalizing_on_cross_docking

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Overview: Network design options

Network Structure	Pros	Cons
Direct shipping	No intermediate warehouse Simple to coordinate	High inventories (due to large lot size) Significant receiving expense
Direct shipping with milk runs	Lower transportation costs for small lots Lower inventories	Increased coordination complexity
All shipments via central DC with inventory storage	Lower inbound transportation cost through consolidation	Increased inventory cost Increased handling at DC
All shipments via central DC with cross-dock	Low inventory requirement Lower transportation cost through consolidation	Increased coordination complexity
Shipping via DC using milk runs	Lower outbound transportation cost for small lots	Further increase in coordination complexity
Tailored network	Transportation choice best matches needs of individual product and store	Highest coordination complexity

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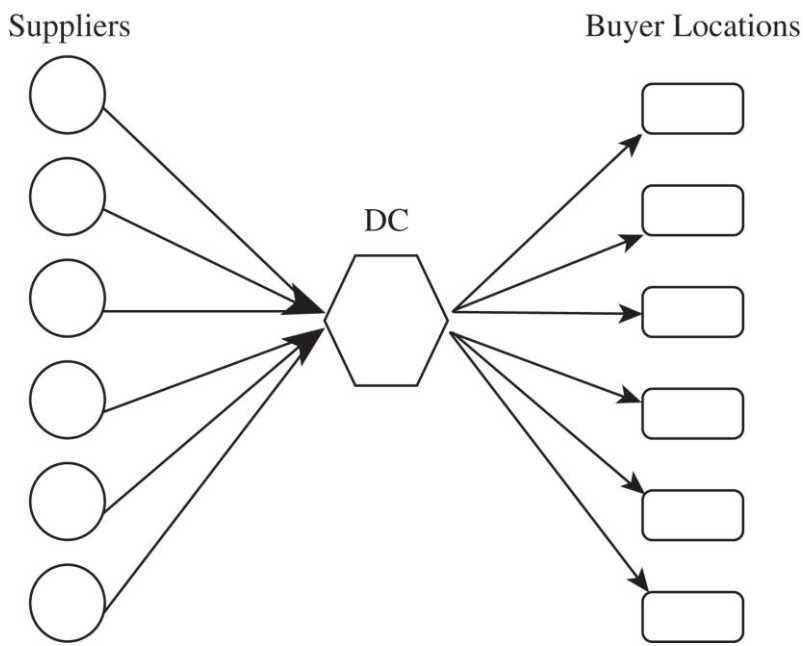
All shipments via intermediate DC with crossdocking

Disadvantages:
Increased coordination complexity
Increased handling at DC



Advantages:
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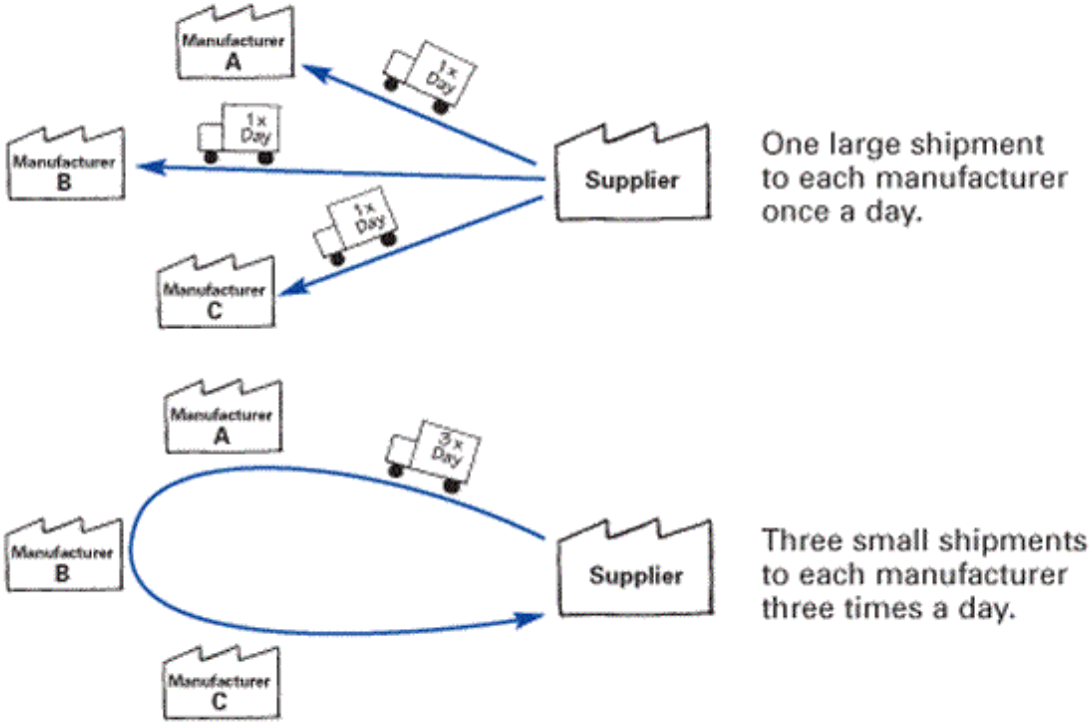
Milk run



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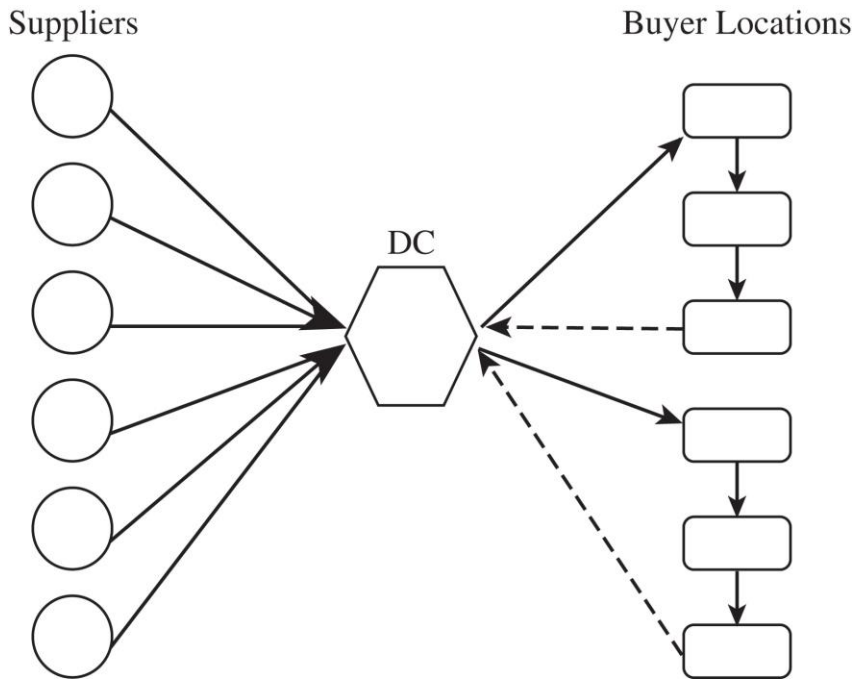
Shipping via DC using milk runs

Disadvantages:
Further increase in coordination complexity



Advantages:
Lower outbound transportation cost for small lots

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COTTON: Planet Money makes a t-shirt

- Part 1
- Part 2
- Part 3

- Take notes to develop a flowchart of the flow of goods around the globe

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The IKEA Group: The story of how we work

- IKEA

- Take notes related to their logistics operations, i.e., how they distribute their goods

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